

MD NAHID HASAN

PhD Student in Computer Science

Charlotte, NC, USA • nahidhasanraju1999@gmail.com • mhasan29@charlotte.edu • +1 (513) 807-5111
nahid-hasan-raju.github.io • github.com/nahid-hasan-raju

RESEARCH INTERESTS

Software Engineering for AI, Large Language Models (LLMs), Artificial Intelligence, Machine Learning, Multi-Agent Systems, Reinforcement Learning, Quantum Computing Applications, Computer Vision, Optimization Systems, Interdisciplinary Computing

EDUCATION

University of North Carolina at Charlotte Charlotte, NC, USA
PhD in Computer Science 2026 – Present

Advisor: Dr. Rrezarta Krasniqi

Research Focus: Software Engineering for AI – applying AI and Large Language Models (LLMs) to improve software engineering

Role: Graduate Teaching Assistant

Miami University Oxford, OH, USA
M.S. in Computer Science Aug 2024 – May 2026

GPA: 4.00/4.00 • Graduate Teaching Assistantship (Full Funding)

Thesis: “A Scalable Hierarchical Multi-Agent System for Urban Delivery Using Q-Learning”

Coursework: Machine Learning, AI, Deep Learning, Advanced ML, Cryptography, Remote Sensing

Rajshahi University Rajshahi, Bangladesh
B.S. in Computer Science & Engineering Jan 2017 – Apr 2022

GPA: 3.78/4.00 • Dean’s List (2018–2020)

Thesis: “Neural Processing of Emotional Musical and Nonmusical Stimuli in Major Depressive Disorder”

Coursework: Data Structures, Algorithms, Linear Algebra, Software Engineering, Database Systems

RESEARCH EXPERIENCE

PhD Research 2026 – Present
University of North Carolina at Charlotte Charlotte, NC, USA

- Advised by **Dr. Rrezarta Krasniqi**; researching applications of **AI and Large Language Models (LLMs)** to improve software engineering practices, workflows, and tooling.
- *[Specific project details to be added]*

MS Thesis Aug 2024 – May 2026
Miami University Oxford, OH, USA

- Designed a scalable **multi-agent urban delivery framework (QuikDel)** using **Q-Reinforcement Learning** for real-time path planning and dispatching.
- Achieved a **25% reduction in travel distance** while sustaining throughput of **13,866+ requests/hour** in large-scale simulations.
- Integrated **GIS-based routing, hierarchical reinforcement learning**, and environment modeling for robust optimization of urban logistics. [\[Details\]](#)

Research Intern May 2025 – Aug 2025
University of Dayton Research Institute (UDRI) Dayton, OH, USA

- Conducted research on **DNA as a computational model for quantum systems**, focusing on electron dynamics and charge transfer mechanisms.
- Applied **real-time time-dependent density functional theory (RT-TDDFT)** and **DFT** to simulate molecular excitation and analyze energy redistribution across nucleobases.

PUBLICATIONS, AND PRESENTATIONS

- M.N. Hasan, R.J. Kilgore, Y. Ozdemir, A. Mehra, A. Mathur, V. Raychoudhury, et al., “QuikDel: A Scalable Hierarchical Multi-agent System for Urban Delivery Using Q-Learning,” *IEEE Access*, 2026.
- “Resilient Multi-Agent Urban Food Delivery System Using a Unified MST Backbone,” *ICAPS*, 2026. **(Under Review)**.
- “QiC-Del: A Cascading Hybrid Quantum-Classical Architecture for Last-mile Delivery,” *IEEE QUANTUM SOFTWARE QSW*, 2026. **(Under Review)**.
- “A Novel and Scalable Distributed Approach for Efficient Long-Haul Delivery,” *IEEE Access*. **(Under Review)**.
- Research presented as posters at multiple venues, including the *Miami AI Symposium 2025*, *AI in Humanities Symposium*, *MUCAT Research Symposium*, *Grad Research Forum*, and *CEC Design Expo*.
- Survey on online delivery optimization techniques. **(In Progress)**

TEACHING EXPERIENCE

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|--|----------------|
| Graduate Teaching Assistant | 2026 – Present |
| <i>University of North Carolina at Charlotte</i> | Charlotte, NC |
- **Course:** ITIS 3300 – Software Requirements, Design, and Testing
 - *[Specific responsibilities to be added]*
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|------------------------------------|---------------------|
| Graduate Teaching Assistant | Aug 2024 – May 2026 |
| <i>Miami University</i> | Oxford, OH |
- Assisted instruction in **Machine Learning, Artificial Intelligence, and Systems Engineering** courses at the graduate and undergraduate level.
 - Mentored **60+ students** through assignments, coding labs, and semester-long projects, emphasizing problem-solving and research practices.
 - Guided **capstone teams and class project groups**, providing feedback on experimental design, implementation, and reporting.

PROFESSIONAL EXPERIENCE

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|--|-------------|
| Machine Learning / Computer Vision Intern | Summer 2026 |
| <i>Innovision LLC</i> | |
- Applied **3D scan processing and computer vision** to help track burn scar healing more objectively over time, using a real clinical dataset of 3D facial scans from burn patients.
 - Built and tested computer vision and deep learning pipelines for real-world 3D medical imaging data; benchmarked segmentation approaches for a specialized healthcare application.
 - Contributed across the project, from early research to development and initial deployment, on a healthcare-focused AI application.
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|---|---------------------|
| Software Engineer / Senior Software Engineer | Jul 2022 – Jul 2024 |
| <i>Samsung Research & Development Institute</i> | Dhaka, Bangladesh |
- Developed and maintained multiple features for Galaxy Watch 6/7 and Fit3 plugins, serving 10M+ users worldwide.
 - Implemented critical functionalities, i.e. Sound Settings, Display Settings, Band Rotate, About Band
 - Collaborated across teams, mentored juniors, and contributed to key architectural decisions.

OTHER RESEARCH PROJECTS

- **3D Scan Processing for Burn Scar Healing Assessment** Applied 3D scan processing and computer vision to objectively track burn scar healing over time using a clinical dataset of 3D facial scans. Built and benchmarked segmentation and deep learning pipelines for 3D medical imaging. (*Innovision LLC, Summer 2026*)

- **Human Activity Recognition** Compared 10 machine learning models (Logistic Regression, Random Forest, Neural Networks, etc.) for activity classification with preprocessing and evaluation. [\[Code\]](#)
- **Analysis of Urban Sprawl using Remote Sensing** Investigated Dhaka City's urban expansion (2013–2023) using Landsat 8 imagery. Applied indices (NDBI, SAVI, MNDWI, IBI) revealing 63.02% urbanization in a 691.51 sq km area. [\[Code\]](#)
- **Neural Processing of Emotional Musical Stimuli** Analyzed fMRI data of depressive vs non-depressive subjects under emotional musical stimuli. Applied PCA + SVM on BOLD signals, achieving best accuracy with Occipital Pole ROI. [\[Code\]](#)
- **Deep Learning-Based Road Skeletonization** Designed U-Net based model for extracting one-pixel road skeletons from noisy imagery. Achieved Dice = 0.87 and IoU = 0.79 on OSM data. [\[Code\]](#)
- **Feistel-Based Cryptographic Algorithm** Implemented custom Feistel cipher with 32-bit key, LFSR-based key scheduling, S-Box transformation, and P-Box permutation. [\[Code\]](#)
- **SoccerBall Detection** Implemented a CNN model to detect soccer balls from field images [\[Code\]](#)

AWARDS AND HONORS

- Graduate Teaching Assistant, Miami University – fully funded scholarship
- Graduate Summer Research Fellowship-2025, Miami University
- Excellence Awards (2×) and Professional Software Certification, Samsung R&D Institute – for outstanding product contributions and engineering expertise.
- Dean's List (3×), Rajshahi University; Perfect GPA 4.00/4.00, Miami University
- Competitive Programming – 1300+ problems solved; Codeforces rating 1544; ICPC Regional rank 74th

LEADERSHIP AND SERVICE

- **Graduate Mentor:** Mentored two capstone teams on quantum-inspired optimization project
- **Social Chair:** Chess Club, Miami University – Organized chess tournaments and events
- **IEEE Events:** Participated in and volunteered on technical workshops and conferences

TECHNICAL SKILLS

- **Programming & Scripting:** Python, C, C++, C#, Java, Kotlin, Unity, JavaScript, Bash, Assembly
- **Machine Learning & AI:** TensorFlow, PyTorch, Scikit-learn, Reinforcement Learning, Deep Learning, Computer Vision, Natural Language Processing
- **Quantum Computing:** PySCF, RT-TDDFT, DFT, Molecular Simulation, Ising Models, QAOA
- **Data Analysis & Visualization:** NumPy, Pandas, Matplotlib, Remote Sensing, fMRI Data Analysis
- **Tools & Platforms:** Git, Linux, LaTeX, Docker, VS Code, Jupyter
- **Other:** Multi-Agent Systems, Optimization Algorithms, Scientific Computing